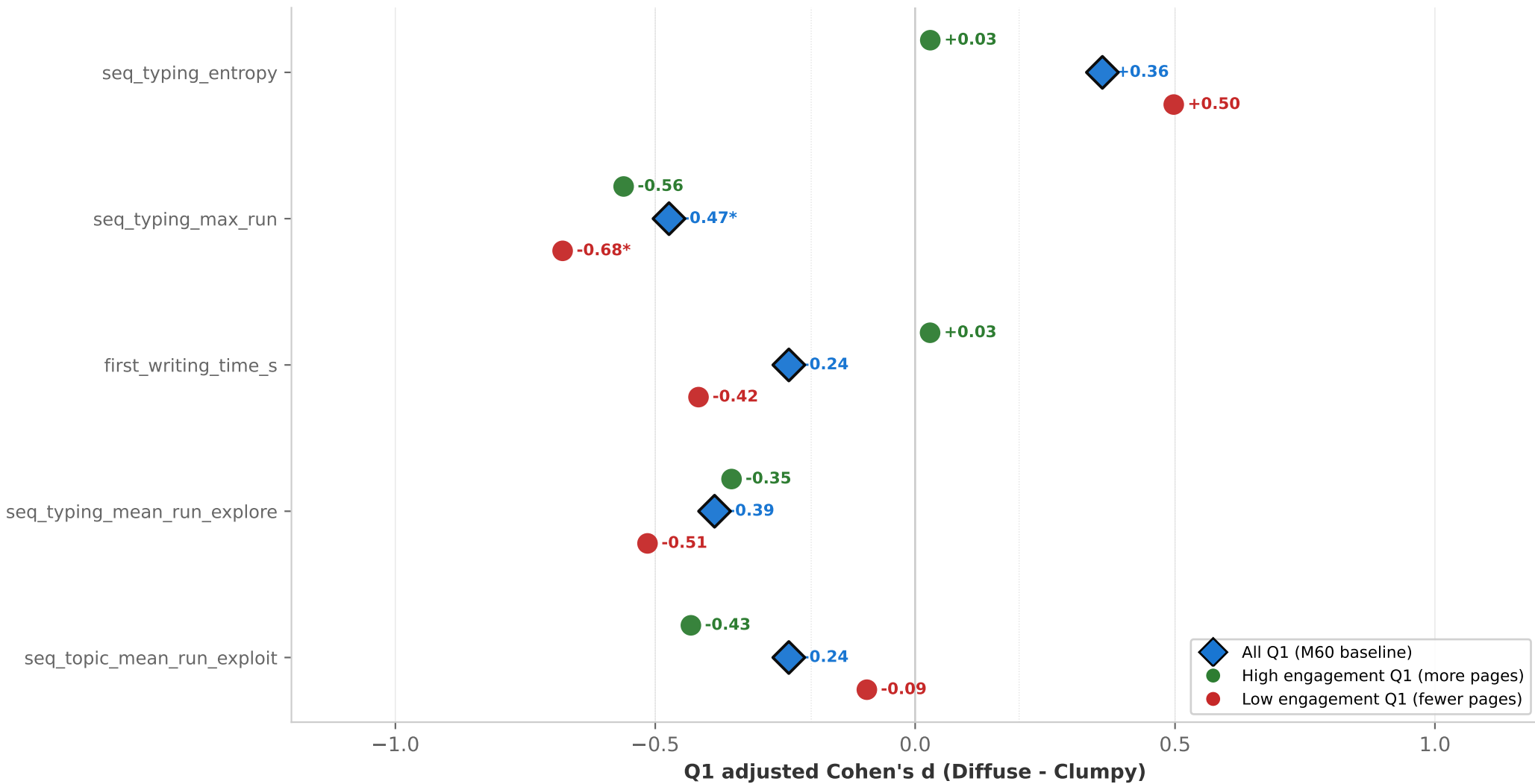


Warm-up test: Q1 effect by Q1 engagement (n_pages median split)



M61: Warm-up Test - Q1 Effect by Q1 Engagement

Engagement split: median Q1 n_pages = 7.0

high_engagement = participants with more than median pages on Q1

low_engagement = participants with at-or-below-median pages on Q1

Hypothesis tested: if warm-up explains M60 (Q2 > Q1), then participants who engaged more with Q1 should already show the condition effect on Q1.

Measure	Group	N	Raw d	Raw p	Adj d	Adj p
seq_typing_entropy	all	90	+0.28	0.191	+0.36	0.097
	high_engagement	35	-0.26	0.465	+0.03	0.935
	low_engagement	55	+0.48	0.090	+0.50	0.074
seq_typing_max_run	all	90	-0.42	0.052	-0.47	0.030
	high_engagement	35	-0.25	0.459	-0.56	0.126
	low_engagement	55	-0.68	0.015	-0.68	0.016
first_writing_time_s	all	90	-0.12	0.567	-0.24	0.261
	high_engagement	35	+0.19	0.590	+0.03	0.936
	low_engagement	55	-0.34	0.228	-0.42	0.132
seq_typing_mean_run_explore	all	90	-0.33	0.137	-0.39	0.084
	high_engagement	35	-0.08	0.818	-0.35	0.330
	low_engagement	55	-0.52	0.076	-0.51	0.077
seq_topic_mean_run_exploit	all	90	-0.19	0.373	-0.24	0.261
	high_engagement	35	-0.41	0.239	-0.43	0.235
	low_engagement	55	-0.05	0.864	-0.09	0.734

Verdict pattern:

If high-eng Q1 effect >> low-eng Q1 effect AND high-eng matches M60-Q2:

-> warm-up hypothesis SUPPORTED

If both groups show similar Q1 effect:

-> warm-up hypothesis NOT supported (effect just mounts over time)

If low-eng Q1 effect > high-eng Q1 effect:

-> contrary - effect might be triggered by under-engagement